deepMerge.test.js U

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```
const assert = require('assert');
const { deepMerge } = require('./deepMerge');

// Test Suite: deepMerge

// Scenario 1: Merging simple (non-nested) properties
function testSimpleMerge() {
  const target = { a: 1, b: 2 };
  const source = { b: 20, c: 30 };
  const result = deepMerge(target, source);

  assert.deepStrictEqual(result, { a: 1, b: 20, c: 30 });
  console.log('✓ Test 1 Passed: Simple non-nested properties merged correctly');
}

// Scenario 2: Merging deeply nested objects
function testDeepNestedMerge() {
  const target = {
    user: {
      name: 'Alice',
      settings: { theme: 'dark', notifications: true }
    },
    version: 1
  };

  const source = {
    user: {
      settings: { theme: 'light', sound: false },
      role: 'admin'
    }
  };

  const result = deepMerge(target, source);

  assert.deepStrictEqual(result, {
    user: {
      name: 'Alice',
      settings: { theme: 'light', notifications: true, sound: false },
      role: 'admin'
    },
    version: 1
  });
  console.log('✓ Test 2 Passed: Deeply nested objects recursively merged without overwriting s
}

// Run tests
try {
  testSimpleMerge();
  testDeepNestedMerge();
  console.log('\nAll unit tests passed successfully!');
} catch (err) {
  console.error('Test failed:', err);
  process.exit(1);
}
```

High Level Programming Overview

done what next

"Generate unit tests for deepMerge covering two distinct scenarios: 1) merging simple non-nested properties, and 2) merging deeply nested objects to verify recursive merging."

Here is the unit test suite for `deepMerge`, covering both simple property merging and deep recursive object merging:

javascript

```
const assert = require('assert');
const { deepMerge } = require('./deepMerge');

describe('deepMerge Quality Assurance Tests', () => {

  // Scenario 1: Merging simple (non-nested) properties
  it('should merge simple top-level properties and overwrite colliding primitives', () => {
    const target = { a: 1, b: 2 };
    const source = { b: 20, c: 30 };

    const result = deepMerge(target, source);

    assert.deepStrictEqual(result, { a: 1, b: 20, c: 30 });
    console.log('✓ Scenario 1 Passed: Simple non-nested properties merged correctly.');
  });

  // Scenario 2: Merging deeply nested objects (proving "deep" capability)
  it('should recursively merge nested objects without overwriting existing sibling keys', () => {
    const target = {
      user: {
        name: 'Alice',
        settings: { theme: 'dark', notifications: true }
      },
      version: 1
    };

    const source = {
      user: {
        settings: { theme: 'light', sound: false },
        role: 'admin'
      }
    };

    const result = deepMerge(target, source);

    assert.deepStrictEqual(result, {
      user: {
```

Review Changes

Ask anything, @ to mention, / for actions

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